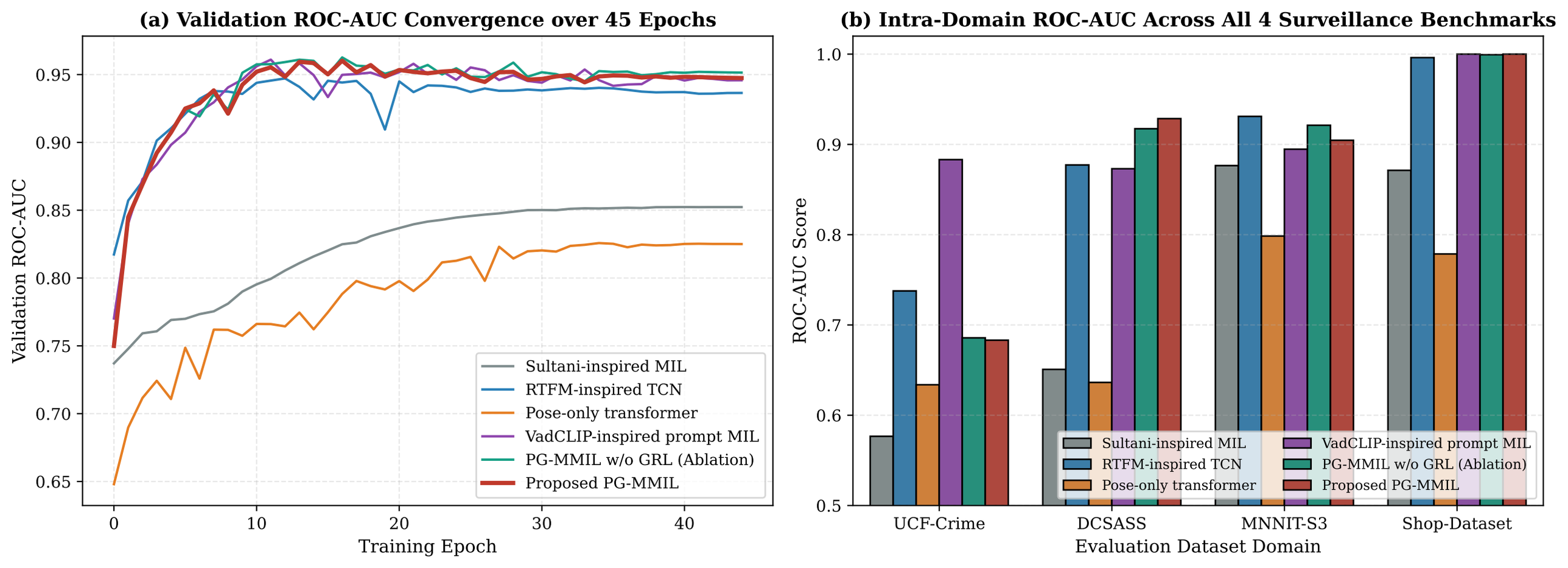
(a) Validation ROC-AUC Convergence over 45 Epochs
Validation ROC-AUC
Training Epoch
0.65
0.70
0.75
0.80
0.85
0.90
0.95
0
10
20
30
40
Sultani-inspired MIL
RTFM-inspired TCN
Pose-only transformer
VadCLIP-inspired prompt MIL
PG-MMIL w/o GRL (Ablation)
Proposed PG-MMIL
(b) Intra-Domain ROC-AUC Across All 4 Surveillance Benchmarks
ROC-AUC Score
Evaluation Dataset Domain
0.5
0.6
0.7
0.8
0.9
1.0
UCF-Crime
DCSASS
MNNIT-S3
Shop-Dataset
Sultani-inspired MIL
RTFM-inspired TCN
Pose-only transformer
VadCLIP-inspired prompt MIL
PG-MMIL w/o GRL (Ablation)
Proposed PG-MMIL